Digital Library Helpdesk Prototype

User Guide for Library Staff

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ACS
2025/26

March 2026

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User Guide

The User Experience | Part A

Understanding the user's journey helps you know exactly what has already happened before a ticket lands in your queue, so you never have to ask a user to repeat themselves.

System Overview

The KdG Library Helpdesk is a chat-based assistant that library users access through their browser. An AI assistant, **named Bib,** handles straightforward questions automatically. When a question is too complex or sensitive, Bib connects the user to a librarian in real time.

What Bib can answer on its own:

- Opening hours (Campus Zuid and Campus Groenplaats)
- Borrowing periods and renewal rules
- Loan limits per campus
- Basic APA referencing guidance
- Remote access to library databases (VPN instructions)
- Inter-library loan procedures
- General account access questions

What always gets escalated to a librarian:

- Blocked accounts
- Fine disputes
- Returns that are not registered in the system
- Research consultations
- Database access problems
- Detailed referencing review requests

What the helpdesk cannot do:

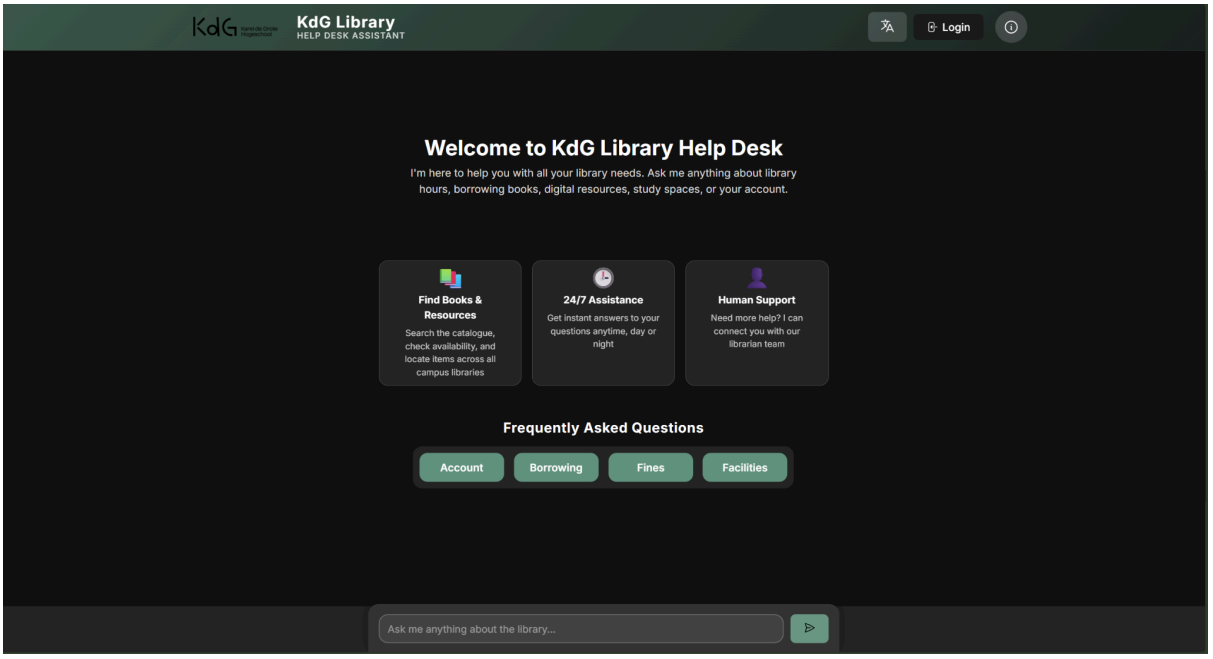
- Access or modify accounts directly
- Confirm real-time book availability
- Make reservations or extend loans
- Answer questions unrelated to the library

How Users Access the Helpdesk

Users reach the helpdesk through their browser. There are two types of users:

Type	Description
Registered user / staff	Has a KdG login account. Sees their chat history and can continue previous conversations.
Guest	Uses the chatbot without logging in. Gets the same AI assistance but must fill in contact details if they need to be connected to a librarian.

The landing page looks like this



Landing page. KdG logo and "Help Desk Assistant" header at top, three feature cards in the middle, an FAQ section below, and a text input bar at the bottom.

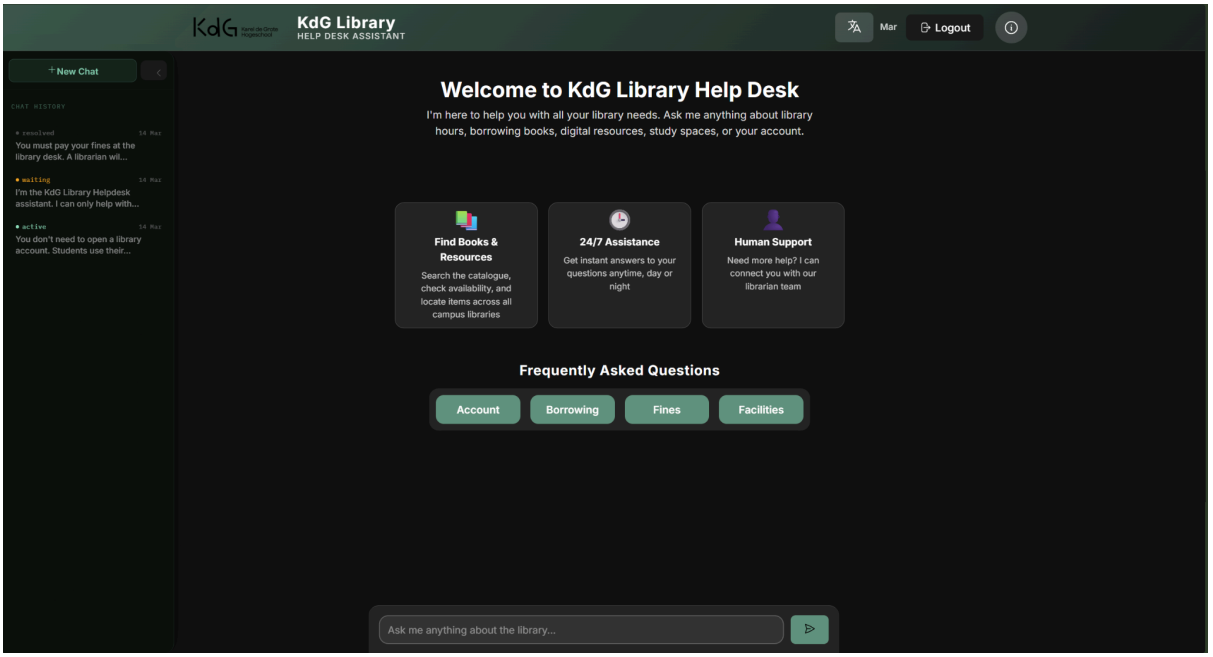
The landing page contains:

- A **welcome text** explaining what Bib can help with
- Three **feature cards** (Find Books & Resources / 24-7 Assistance / Human Support)
- A **Frequently Asked Questions** panel with expandable topics (Account, Borrowing, Fines, Facilities)
- A **chat input** bar at the bottom with the placeholder: "Ask me anything about the library..."

Users can start a chat either by:

- Typing their question in the input bar and pressing Enter or the send button
- Clicking any of the ready-made FAQ questions, which send that question automatically

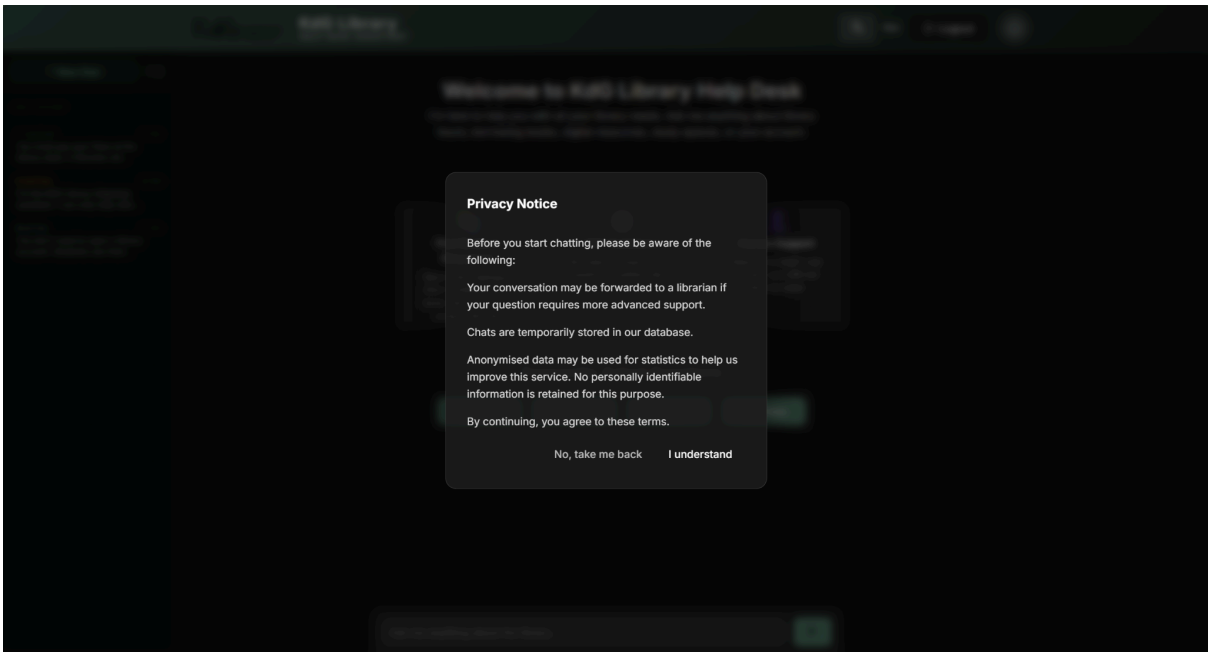
Registered users also see a sidebar on the left with their chat history, a list of past conversations with date and status. They can return to any previous chat or start a new one.



Landing page for a logged-in user. Left sidebar with "New Chat" button and list of previous conversations, main area with hero text and input bar.

The Privacy Notice

Before any user can start chatting, on every new browser session, the application shows a privacy notice modal. This notice blocks the entire screen until the user actively dismisses it.



Privacy notice modal. Dark overlay covering the full screen, white text with four bullet points, "No, take me back" and "I understand" buttons at the bottom

The notice contains the following information

"Before you start chatting, please be aware of the following:

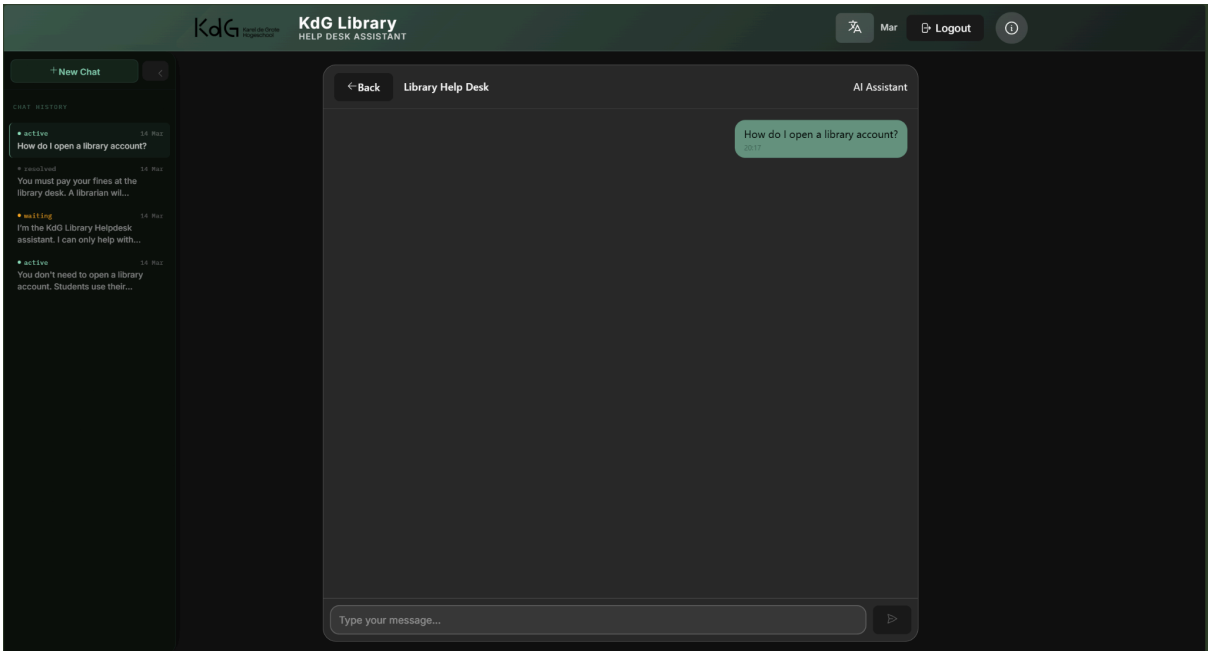
Your conversation may be forwarded to a librarian if your question requires more advanced support.
Chats are temporarily stored in our database.
Anonymised data may be used for statistics to help us improve this service. No personally identifiable information is retained for this purpose.
By continuing, you agree to these terms."

Two buttons are available

- **"No, take me back"**
  - Returns the user to the KdG website (kdg.be)
- **"I understand"**
  - Dismisses the notice and shows the helpdesk

The Chat Interface | What users See

Once the user types a question and presses send, the screen switches to the chat view.



user chat screen. Top bar with "Back" button and status chip, scrollable message area in the middle, text input and send button at the bottom.

Top bar:

- A **Back** button (arrow icon) to return to the landing page
- The title: "**Library Help Desk**"
- A **status chip** on the right showing the current state of the conversation (see table below)

Status chips the user sees:

Chip	Meaning
AI Assistant	Bib is handling the conversation
Escalation	The AI has suggested connecting to a librarian
Waiting for librarian	The user accepted escalation and is waiting for a librarian
Librarian	A librarian is in the chat
Resolved	The conversation has been closed

Message bubbles:

- The user's own messages appear in light grey on the right
- AI responses appear in light gray on the left
- Librarian messages appear in light blue on the left
- System updates (e.g., "A librarian has joined the chat.") appear centred in italics

Every message shows a timestamp in the corner (e.g., 14:30).

How the AI Handles a Question

When a user sends a message, Bib searches its knowledge base for relevant information, determines what kind of question is being asked, and decides how to respond.

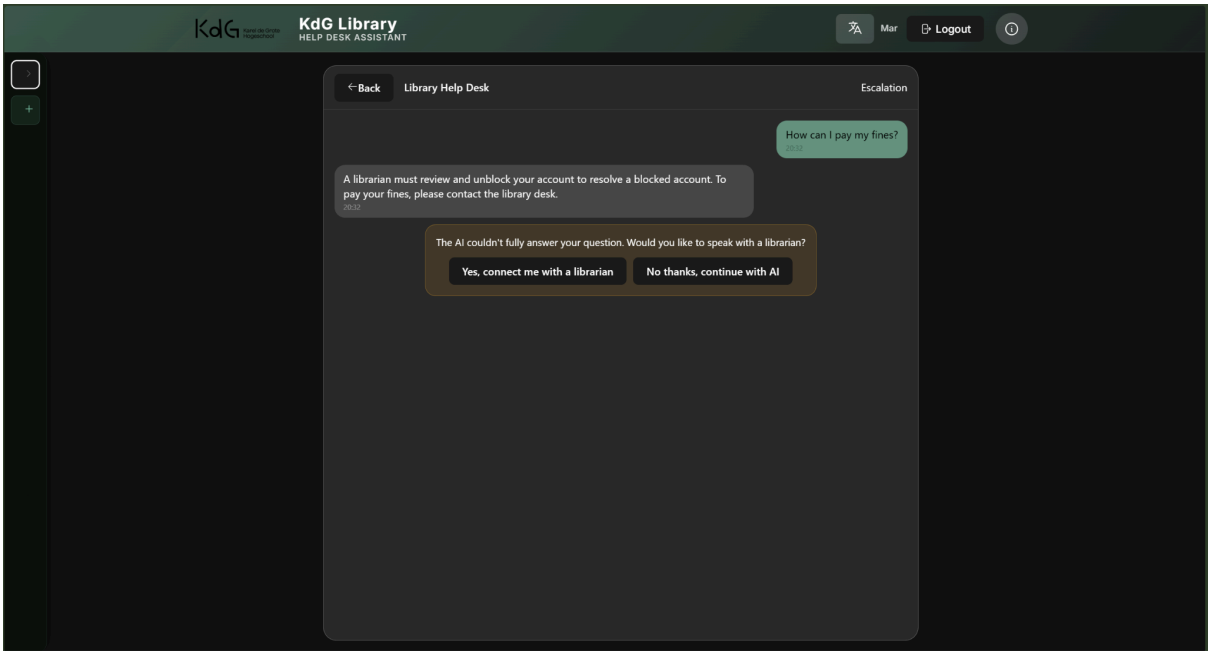
Possible outcomes:

1. **Direct answer (most common):** Bib answers the question and the user can ask follow-up questions.
2. **Follow-up question:** If the answer depends on information Bib does not have yet (for example, which campus a user is at to give accurate loan limits), Bib asks a clarifying question before answering.
    a. *Example: "Which campus are you at — Zuid or Groenplaats?"*
3. **Escalation suggestion:** If the question falls into a category that requires a librarian, or if Bib cannot answer with sufficient confidence, it suggests escalation. The user is not forced to accept.
4. **Out of scope:** If the question is not related to the library at all, Bib politely says it cannot help with that topic.

Escalation | When the AI Connects a User to You

Escalation is the process through which a user is transferred from the AI to a librarian. This is triggered automatically based on the type and complexity of the question.

What the User Sees During Escalation



Chat screen with escalation card. A styled message box with text asking if the user wants to speak to a librarian, and two buttons: "Yes, connect me with a librarian" and "No thanks, continue with AI".

When escalation is triggered, a special card appears in the chat

"The AI couldn't fully answer your question. Would you like to speak with a librarian?"

Two buttons appear

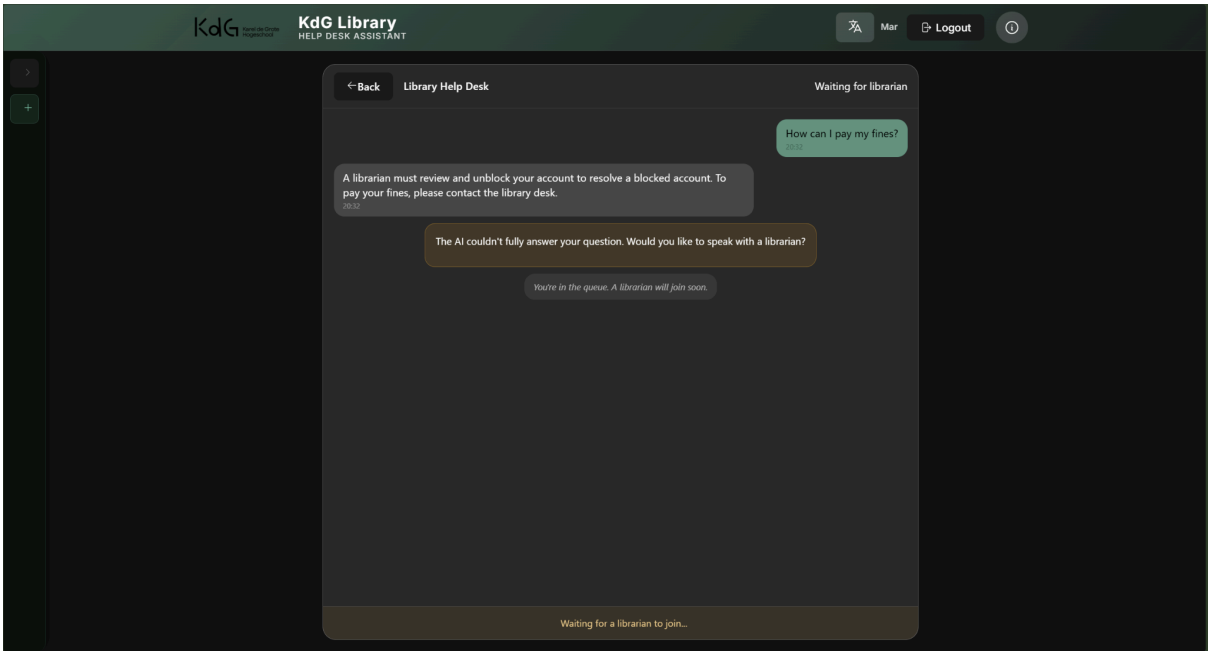
- **"Yes, connect me with a librarian"**
  - The user accepts
- **"No thanks, continue with AI"**
  - The user declines

If the user declines:

- The escalation card disappears
- The chat continues with Bib
- The status chip returns to "AI Assistant"
- No ticket is created

If the user accepts:

1. A brief spinner appears with the text: "Connecting you to a librarian..."
2. The status chip changes to "Waiting for librarian"
3. The input field is disabled, the user waits
4. A ticket is created and appears in your Ticket Queue
5. You receive a notification in Microsoft Teams



Chat screen in waiting state. "Waiting for librarian" chip visible, spinner in the message area, input bar disabled with text "Waiting for a librarian to join..."

When you open the chat:

- The user's status chip changes to "Librarian"
- The user sees the system message: "A librarian has joined the chat."
- The input field re-enables for the user
- The conversation continues in real time between you and the user

When you close the ticket:

- The user sees the banner: "This chat has been resolved. You can start a new chat from the home page."
- The status chip changes to "Resolved"
- The input field is disabled, the conversation is over

Guest Users (Without a Login)

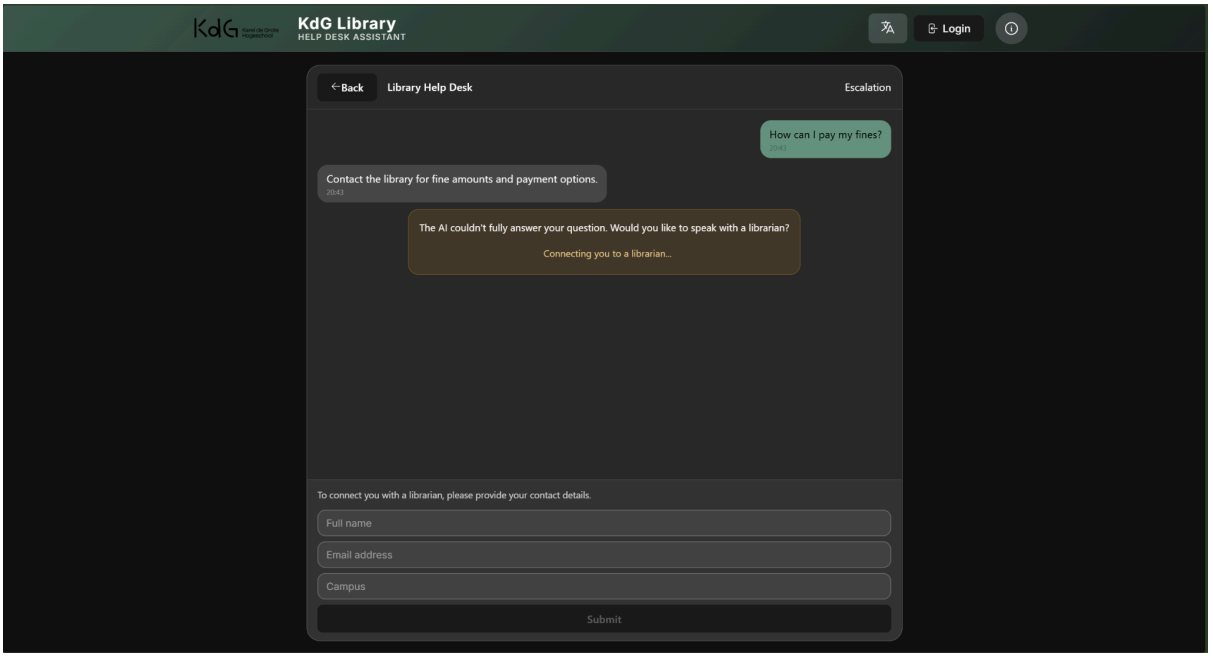
A guest is a user who uses the helpdesk without logging in. They get the same AI experience as a registered user, with one important difference: if they need to be connected to a librarian, the system needs their contact details first.

Guest Chat Start

Guests access the helpdesk the same way as registered users, they accept the privacy notice, see the landing page, and start chatting. No account is needed.

Guest Escalation & Contact Form

When Bib suggests escalation and the guest clicks "Yes, connect me with a librarian", a contact form appears in the chat area before the escalation proceeds.



Guest escalation form. Three fields (Full name, Email address, Campus) with a "Submit" button, appearing where the chat input normally is.

The form has three required fields

- Full name
- Email address
- Campus

The guest must fill in all three fields and click Submit before the escalation continues. If any field is empty, the Submit button remains disabled.

After submission, the escalation proceeds exactly as it does for registered users, the ticket is created, the user enters the waiting state, and you receive a Teams notification.

Guest Resume Link

Because guests are not logged in, they cannot simply return to their chat by revisiting the website. To allow them to continue a conversation (for example, if they need to step away while waiting), the system sends them an email with a unique resume link.

The email contains:

- A short explanation that their request has been received
- A unique link (example format: https://app.kdg.be/chat/resume/[token])

When the guest clicks the link:

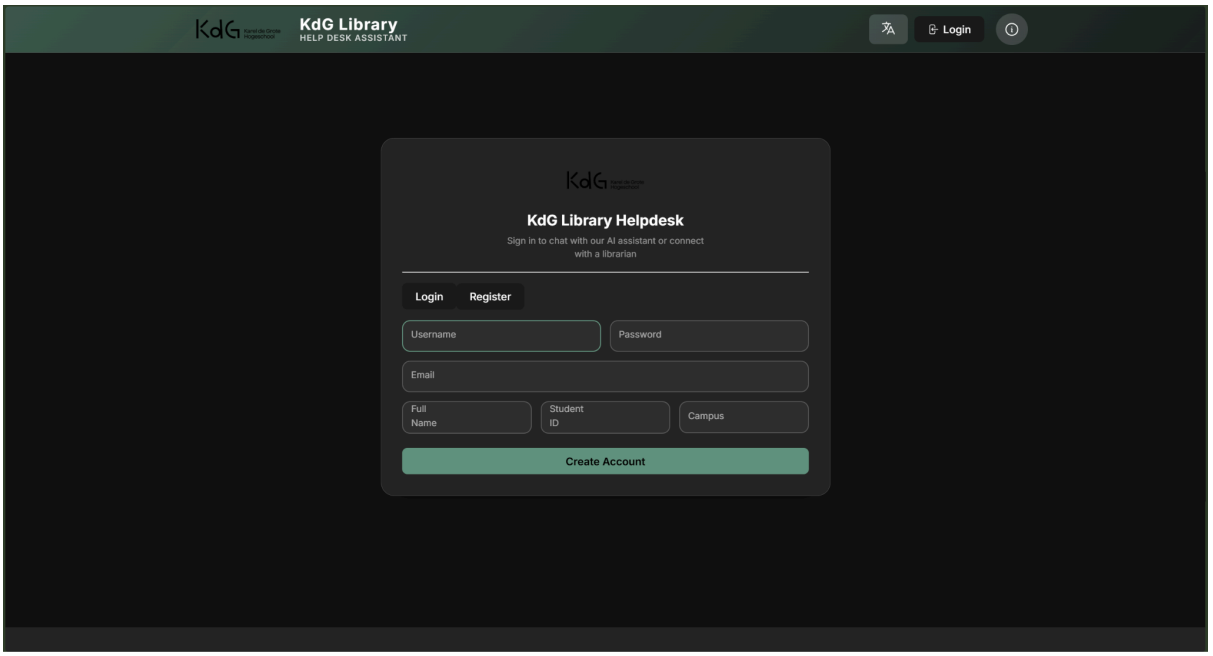
1. The browser opens the helpdesk

2. The system validates the link
3. The guest is returned directly to their conversation

If the link is invalid or has expired, the guest sees: "This resume link is invalid or has expired."
In this case, the guest would need to start a new chat.

The Librarian's Workflow | Part B

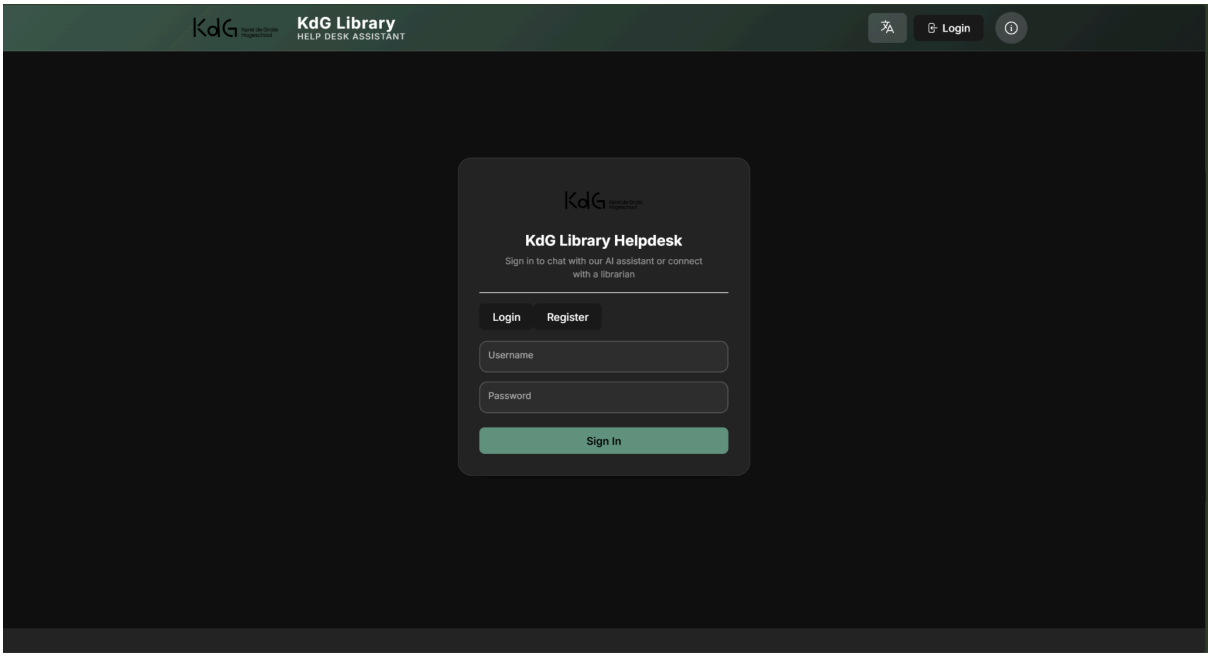
Registration



Registration page. "KdG Library Helpdesk" title, subtitle text, two tabs (Login and Register), username, password, email, full name, user ID, campus fields, and "Create Account" button

Logging in

Open the helpdesk in your browser. Click the Login button in the top right corner, or navigate directly to the login page.



Login page. "KdG Library Helpdesk" title, subtitle text, two tabs (Login and Register), username and password fields, and "Sign In" button

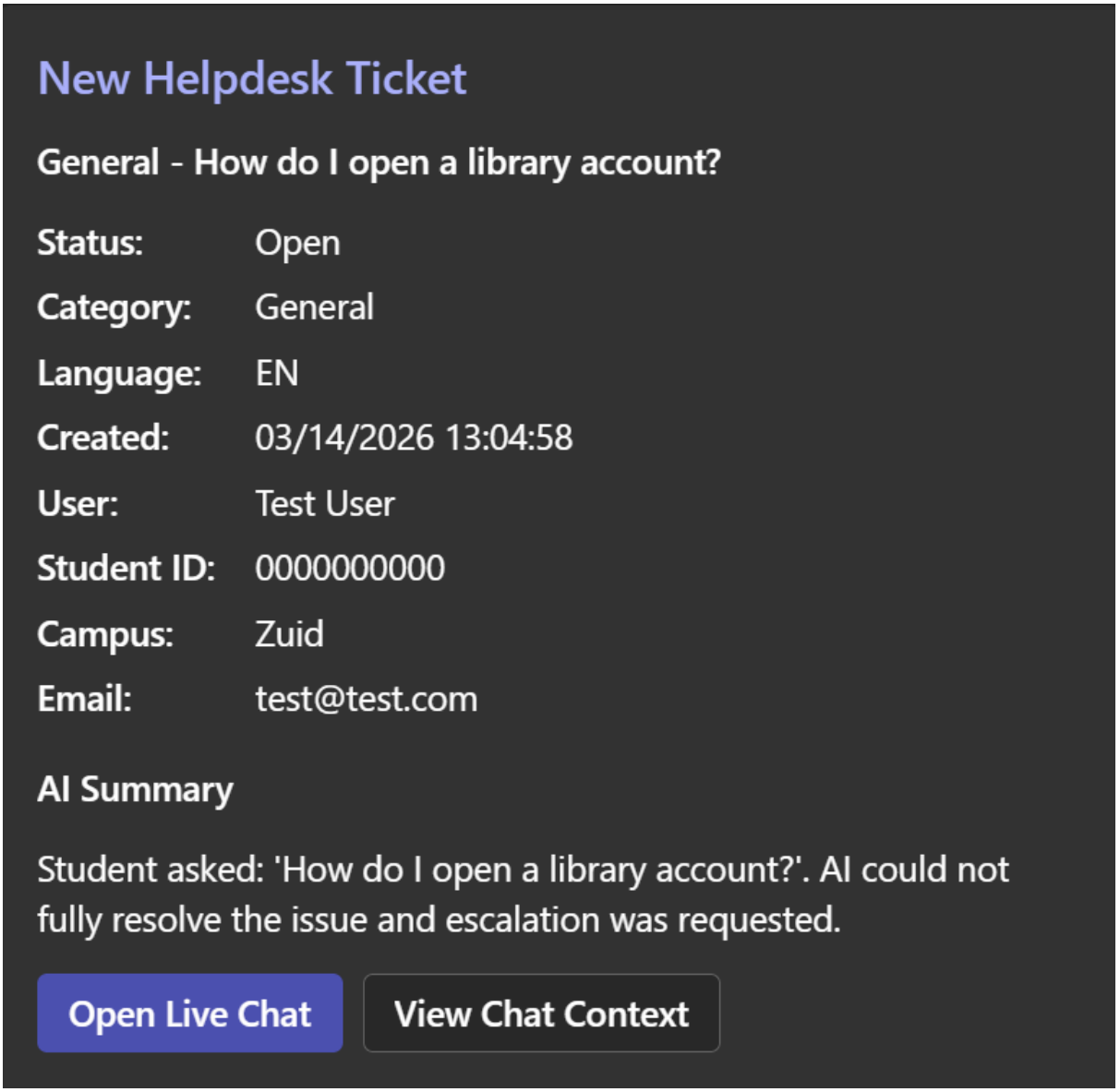
Steps

1. Click the **Login** tab
2. Enter your **username**
3. Enter your **password**
4. Click **Sign In**

After logging in, you are taken directly to the Ticket Queue.
If your credentials are incorrect, the message "Something went wrong. Please try again." appears in red below the form. Check your username and password and try again.

Understanding a Ticket Card

Each ticket appears as a card. Here is a breakdown of what each element means:



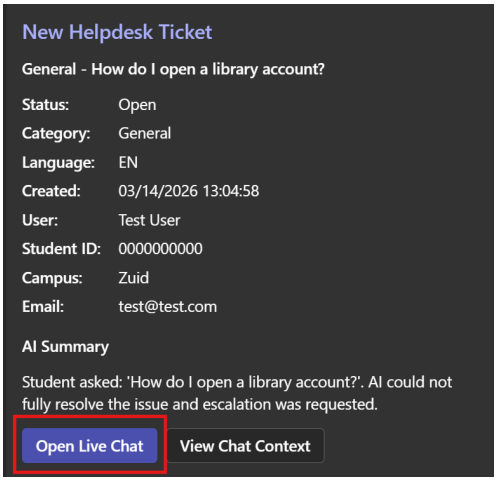
An adaptive card in the teams channel. Title, status, category, language, created timestamp, username, user ID, campus, email, AI summary, and two buttons: "Open Live Chat" and "View Chat Context".

Element	What it tells you
Title	Auto-generated from the conversation and gives you the topic at a glance
Status	Current state: Open or Closed
Category	The type of question: Lending, Account, Research, Academic Writing, Open Hours, General
Language	Whether the user chatted in English (EN) or Dutch (NL)
Created	When the ticket was created
User	Users username

User ID	User ID (staff / user)
Campus	Users main campus
Email	Users email
AI Summary	A brief AI-generated description of the user's problem
Open Live Chat	Click this to enter the live chat with this user
View Chat Context	Click this to see full chat context before escalation

Opening a Chat with a user

Click the **Open Live Chat** button on any ticket card.



What happens immediately when you open a chat:

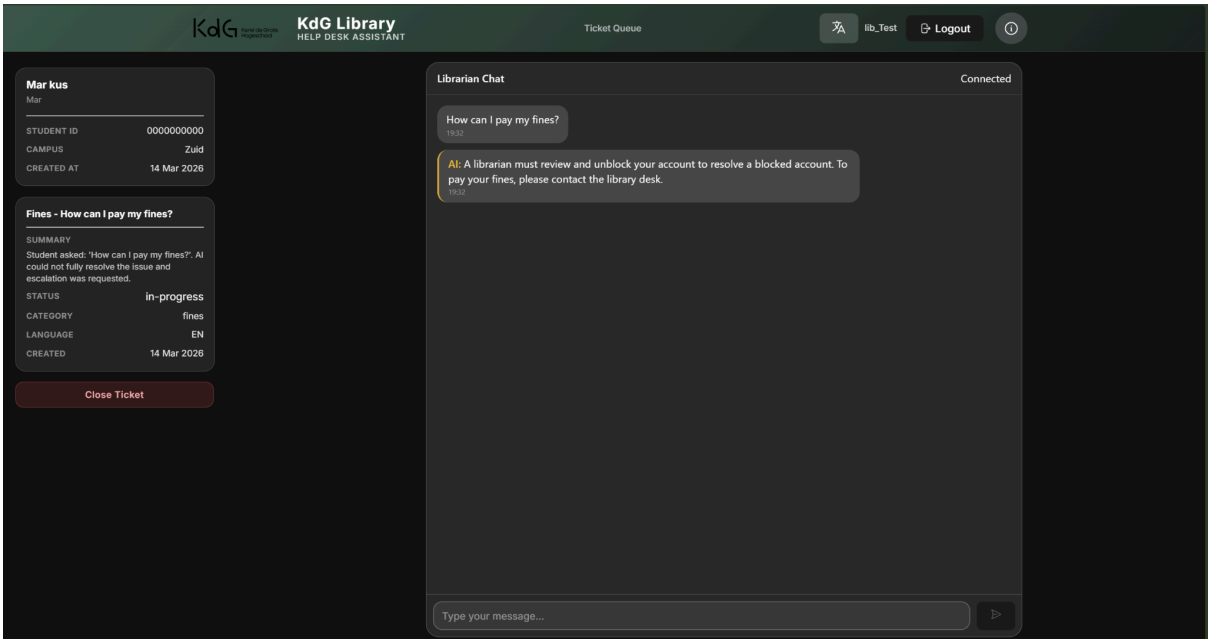
1. The ticket status automatically changes from **Open** to **In Progress** (you do not need to manually pick it up)
2. The user's waiting status changes and they see: "A librarian has joined the chat."
3. The user's input field re-enables and they can type again
4. You see the full conversation history, every message between the user and Bib before escalation
5. You are ready to type

Important

Read the chat context before you type. The AI-generated summary gives you the short version, but the full transcript shows you exactly what was said and what the AI already tried. You will not need to ask the user to explain their question again.

The user is notified the moment you open the chat. Try not to open a chat unless you are ready to respond promptly.

The Chat Screen



Full chat screen. Left sidebar (340px wide) with user Information card and Ticket Details card, right side with message thread and input field at the bottom

The chat screen has two panels side by side.

Left Sidebar | user & Ticket Information

User Information card (top):

Field	What it shows
Name	User's full name
Email	User's email address
User ID	Users number (staff ID / user ID)
Campus	Campus (or "N/A" if not provided)
Created at	Date the ticket was created

Ticket Details card (below):

Field	What it shows
Title	The ticket title
Summary	AI-generated summary of the issue
Status	Current status chip

Category	Question category
Language	EN or NL
Created	Creation date

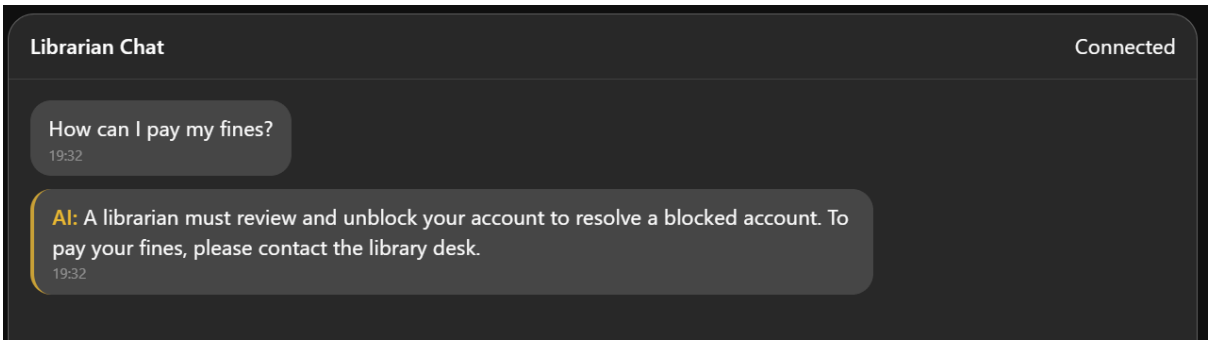
Close Ticket button (bottom of sidebar):

A red **Close Ticket** button appears when the ticket is still open or in progress. This is what you click when the issue is resolved. See Section "Closing a Ticket" for details.

Right Side | The Chat

Connection indicator: In the top right corner of the chat area, a chip shows your connection state:

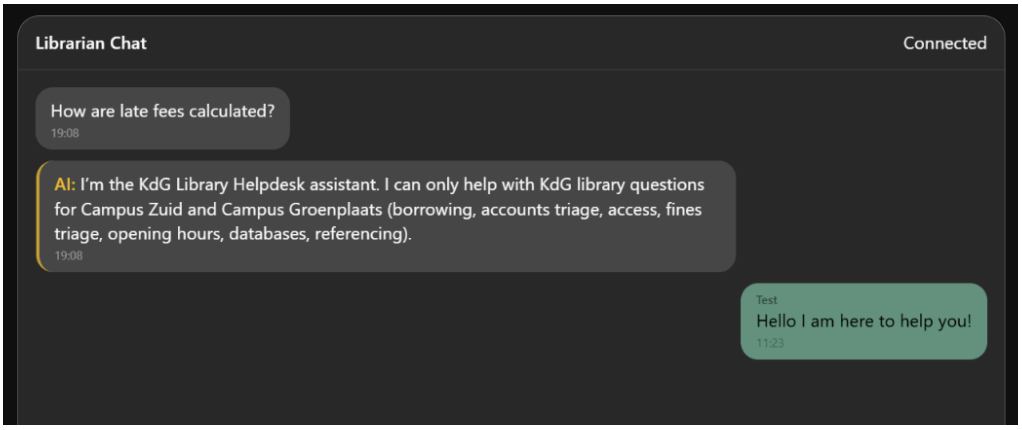
- **"Connected"**
  - You are live, messages reach the user instantly
- **"Resolved"**
  - The ticket has been closed



Chat header with "Connected" chip visible in the top right corner

Reading the Conversation History

When you open the chat, all messages before your arrival are visible. The visual style tells you at a glance who said what:



Appearance	Who sent it
Gray bubble, left side	user
Gray bubble, with yellow left border, labelled “AI”	Bib (The AI assistant) - what Bib told the user
Green bubble, right side	You (the librarian)

Each message shows a timestamp in the format HH:MM (e.g., 14:30).

Sending a Message

Input bar at the bottom of the chat. Text field with placeholder "Type your message...", send button (paper airplane) to the right.

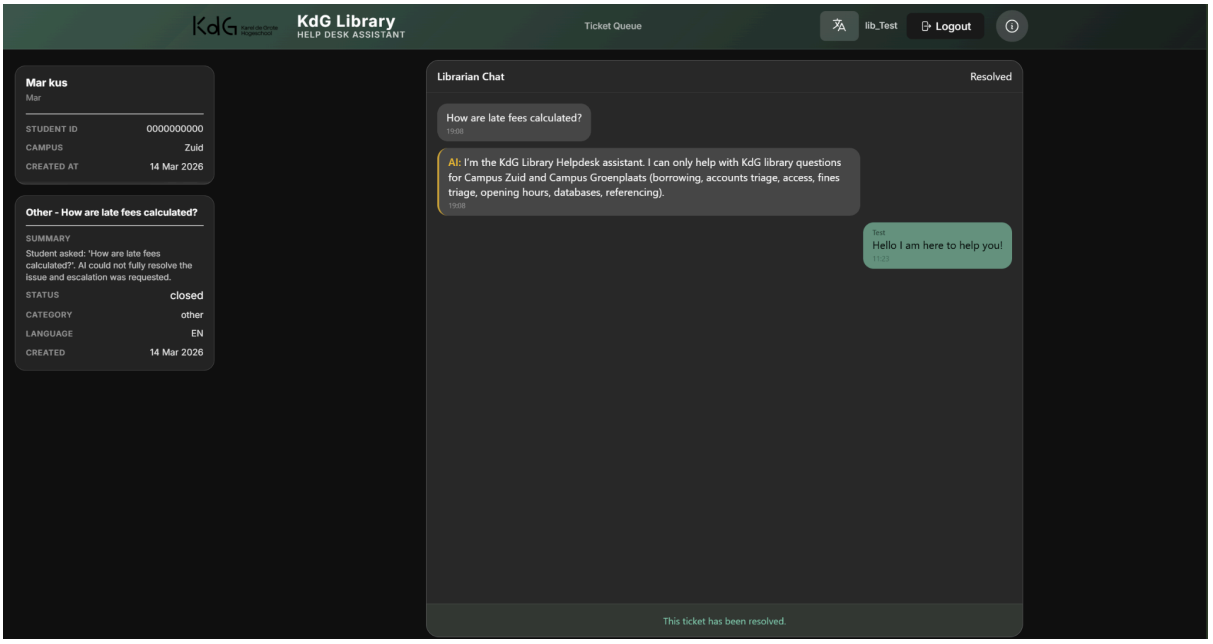
1. Click in the text field (placeholder: "Type your message...")
2. Type your message
3. Press Enter on your keyboard, or click the Send button (paper airplane icon)

Your message appears in green on the right. The user sees it immediately. There is no character limit. You can write multi-line messages.

When the Chat Is Resolved

After you close a ticket, the chat becomes read-only:

- The input field is disabled
- The status chip changes to "Resolved"
- A banner appears: "This ticket has been resolved."
- The user sees: "This chat has been resolved. You can start a new chat from the home page."

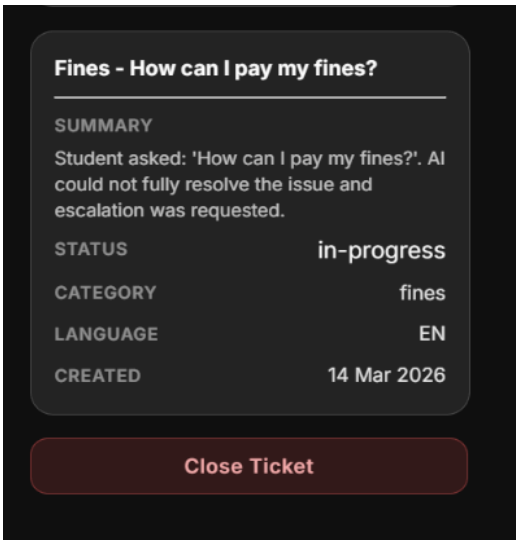


Resolved chat. "Resolved" chip in header, grey "This ticket has been resolved." banner at the bottom where the input used to be

Closing a Ticket

Close a ticket when:

- The user's issue has been fully resolved
- You have provided all the help you can and the user confirms they are satisfied
- The user has stopped responding and is unlikely to return



Left sidebar. Red "Close Ticket" button at the bottom, below the ticket details card.

Steps

1. Confirm the issue is resolved (tell the user something like "Is there anything else I can help you with?")
2. In the left sidebar, click the red Close Ticket button
3. The ticket is closed immediately

What happens automatically

- Ticket status changes to Closed
- The user's chat shows the resolved banner
- Microsoft Lists is updated: the ticket status becomes "Closed"
- No further messages can be sent by either party

Important

There is no confirmation step. Clicking **Close Ticket** acts immediately. If you close a ticket by mistake and the user still needs help, they will need to start a new chat from the helpdesk. Their conversation history from the closed ticket will still be visible if they log in.

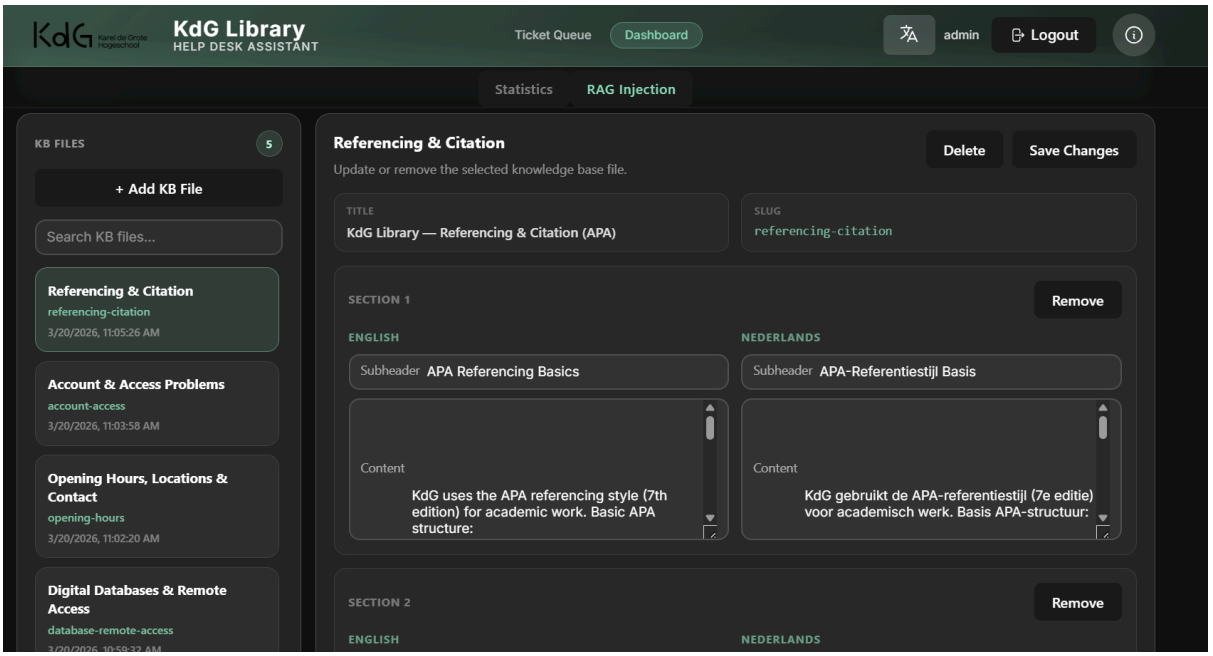
The Admin’s workflow | Part C

RAG Injection | Managing the AI Knowledge Base

What is a Knowledge Base File?

A knowledge base file is a structured document that contains information the AI can use to answer questions.
Each file represents one topic (for example: opening hours, referencing rules, account issues) and is divided into sections using clear headings. The system processes these files and breaks them into smaller chunks that the AI can search through when responding to users.

The RAG Injection Interface



The RAG Injection page consists of two main areas:

- Left side:
  - A list of all existing knowledge base files
  - A search bar to quickly find a file
  - A button to add a new file
- Right side:
  - The editor for the selected file
  - Fields to create, update, or delete content

Creating a New Knowledge Base File

To add new information for the AI:

1. Click “+ Add KB File”
2. Enter a Title
3. The system automatically generates a Slug
4. Add your content in the editor
5. Click Save Changes

Title and Slug

- Title
  - The human-readable name of the file
    - Example: Opening Hours
- Slug
  - A system-friendly identifier
  - Automatically generated from the title
    - Example: opening-hours

The slug can be edited manually if needed, but this is usually not required.

Writing Content for the AI

The content you write is what the AI will use to answer questions. It is important that this content is clear, structured, and easy to understand.

Each file is divided into sections with:

- A subheader
- A content block

You can create multiple sections within one file.

Example Structure

- Subheader:
    - Opening hours for Campus Zuid
- Content:
    - The library at Campus Zuid is open Monday to Friday from 08:30 to 22:00. During exam periods, extended hours may apply.

Best Practices for Writing Content

- Keep information clear and concise
- Use simple and direct language
- Avoid unnecessary details
- Write complete sentences
- Focus on one topic per section

This helps the AI retrieve the correct information more accurately.

Editing Existing Files

To update information:

1. Select a file from the list
2. Modify the content
3. Click Save Changes

Changes are immediately reflected in how the AI answers questions.

Deleting a File

To remove a knowledge base file:

1. Select the file
2. Click Delete

This removes the information from the AI's knowledge base. The AI will no longer use this content in its responses.

How the System Uses This Content

When a user asks a question:

1. The AI searches through all knowledge base files
2. It finds the most relevant sections
3. It uses that information to generate a response

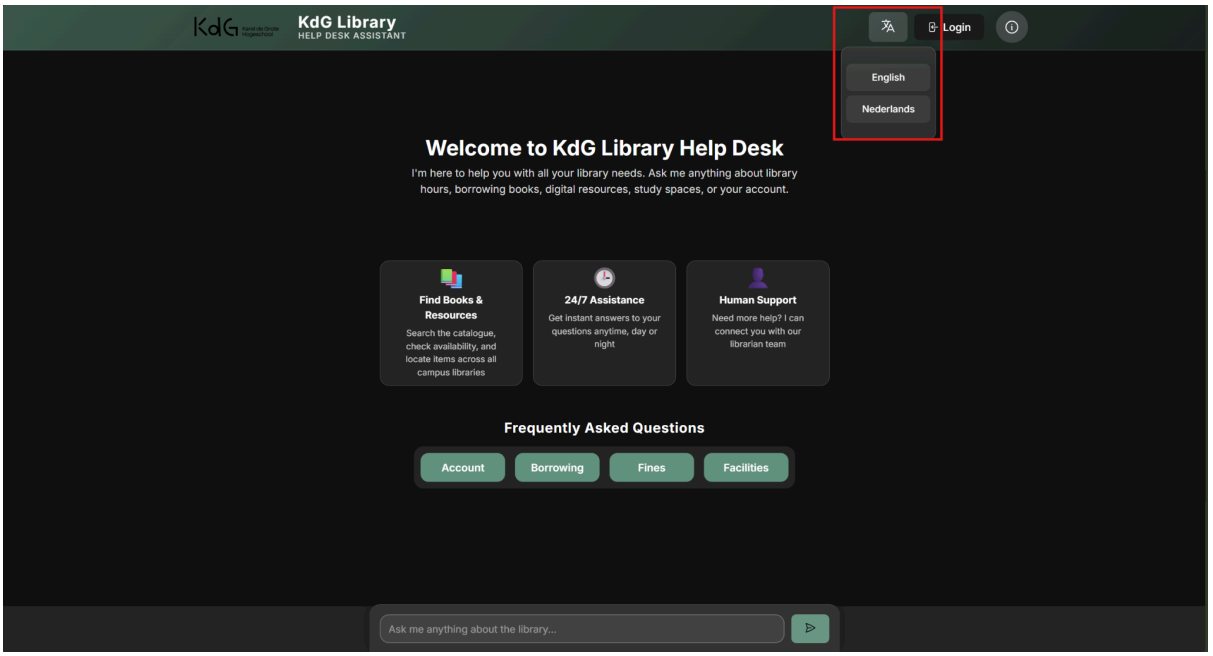
This process ensures that answers are based on the latest information entered by librarians.

Important

The AI can only answer based on the information provided here. If information is missing or unclear, the AI may give incomplete answers.Keeping the knowledge base up to date is essential for correct responses

General | Part D

The interface is fully available in both English and Dutch.



Top navigation bar. Language icon on the right side, with a dropdown showing "English" and "Nederlands".

To switch

1. Click the **language icon** in the top right of the navigation bar
2. Select **English** or **Nederlands**

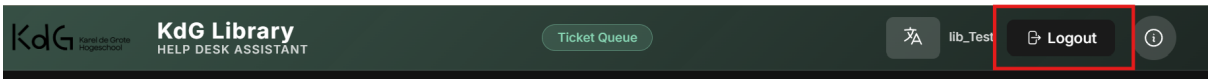
The entire interface switches immediately — all labels, buttons, error messages, and navigation items change language.

Important

Changing the interface language does not affect the user's chat. A user may write in Dutch while you use the English interface. The user's language only affects what Bib writes back to them.

Logging Out

Click the Logout button in the top right of the navigation bar (next to your username).



Top navigation bar. Username visible, Logout button immediately to its right

You are returned to the helpdesk landing page. Always log out when you are done, especially on shared computers.

Error Messages | What They Mean and What To Do

In the Ticket Queue

- **"Failed to load ticket data."**. The system could not retrieve your tickets. Usually a temporary network issue. → Refresh the page. If it continues, check your connection or contact IT.
- **"No tickets in this category."**. Not an error, it simply means no tickets match the selected filter at this moment. → Try the Active filter to see all open work.

In the Chat Screen

- **"No ticket found for this chat."** The ticket for this chat could not be found. It may have been closed by a colleague already. → Return to Team to check the status.
- **Spinner that does not stop.** The system is connecting or loading. Wait up to 15 seconds. → If it persists, refresh the page and open the chat again.
- **No "Connected" chip visible.** Your WebSocket connection is still being established. → Wait a few seconds. If it does not appear, refresh the page.
- **"Failed to close the ticket. It may have already been resolved."** The ticket was already closed. Perhaps you double-clicked, or a colleague resolved it. → Refresh the page. The ticket should now show as Closed. No further action needed.

At Login

- **"Something went wrong. Please try again."** Your username or password is incorrect. → Check your credentials and try again. If you are certain your details are correct, contact your administrator as your account may need to be reset.

General Browser Issues

Symptom	What to try
Page look broken or blank	Press F5 to refresh
Buttons are not responding	Refresh the page, then try again
Changes not saving	Check you internet connection
Page loads very slowly	This may be a server startup delay. Wait 30 sec and refresh

Frequently Asked Questions

Q: A user stopped responding mid-conversation. What should I do?
If you have answered the question and the user is no longer responding, wait a reasonable amount of time (a few minutes) and then close the ticket. The user's chat history is saved, and if they return later and start a new chat, they can see the previous answers.

Q: I need to look something up before answering. Can I leave and come back?
Yes. You can navigate away from the chat page and return using the Ticket Queue. When you leave, your WebSocket connection closes and the user sees: "The librarian has left the chat." When you return and reopen the chat, the user sees: "A librarian has joined the chat." The full conversation history remains intact.

Q: Two of us are working at the same time. How do we know who takes which ticket?
The system does not assign tickets automatically. When a librarian opens a ticket, its status immediately changes to In Progress. This is visible to all librarians in the queue. Use this as your signal that a colleague has taken a ticket. Agree with your colleagues on a workflow (e.g., the librarian who spots it first takes it).

Q: Can I see tickets that were handled yesterday or last week?
Closed tickets are kept in Microsoft Lists, where you can view full history, filter by date, and see all user details and AI summaries. In the helpdesk interface itself, there is currently no closed-tickets filter, as the queue focuses on active work.

Q: A user says the AI gave them incorrect information. What should I do?
Correct the user directly in the chat. You can see the exact AI message in the conversation history (gray bubble with yellow border and the "AI:" label). Provide accurate information. If this seems to be a recurring issue with the same type of question, note it and report it to the system administrator, as the AI knowledge base can be updated.

Q: The user is writing in Dutch, but I would prefer to use English. Is that a problem?

No. The two languages are entirely independent. Your interface language and the user's chat language do not affect each other. However, responding in the user's language where possible improves the experience for them.

Q: Can I send a file or image to a user?

No. The current version of the chat only supports text messages. If you need to share a document (e.g., an instruction guide), paste the relevant information as text or instruct the user to visit a specific web page.

Q: A guest user started a chat and then disappeared. Will they lose their conversation?

No. When a guest escalates, they are emailed a unique resume link. They can click the link at any time to return to their chat. If they have not returned after a reasonable time and you have provided a full answer in the chat, close the ticket, and they will be able to read your answer when they use the resume link.

Q: I accidentally opened a ticket but am not ready to respond yet. What should I do?

Opening a ticket changes its status to In Progress and notifies the user that a librarian has joined. If you navigate away immediately, the user sees: "The librarian has left the chat." This is not ideal but not a serious problem. If possible, send a brief message first: "Hello, I'll be with you in just a moment."

Q: Can users see my name?

No. Your librarian username is displayed above your messages, but only you and other librarians can see it. For the user, in the user's chat view, your name is not displayed.

Q: A ticket was accidentally closed. Can it be reopened?

No. Closed tickets cannot be reopened. If a user still needs help, they must start a new chat from the helpdesk. Their previous conversation history remains accessible to them if they are a registered user.